Reference-Conditioned Structural Assessment of Low-Resource Chinese Calligraphy with Overlapping Stroke Parsing and Human-Audited Spatial Scoring

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Abstract

Computer-vision systems for handwriting commonly assume mutually exclusive semantic regions, optimize recognition or style classification, or compare images without separating global placement from local structural error. These assumptions are unsuitable for Chinese calligraphy, where adhered and crossing strokes may share pixels and where flexible registration can erase the discrepancy that an assessment system should preserve. We formulate low-resource reference-conditioned structural assessment as an auditable document-analysis problem. A six-channel parser predicts five independently activated direction masks and an endpoint mask; U-Net, DeepLabV3+, and SegFormer-B2 are compared with three seeds on quality-controlled standard and character-disjoint splits. A constrained global registration is evaluated on 200 real references using controlled nuisance and structural perturbations. Human association is studied on 150 natural same-character pairs rated blindly by three adults, and aligned spatial-distribution similarity (ASDS) is reported as a retrospective, character-grouped internal-validation analysis. DeepLabV3+ reaches $\mathbf{0.9630 \pm 0.0006}$ standard-split Macro Dice, whereas SegFormer-B2 leads character-disjoint evaluation at $\mathbf{0.8866 \pm 0.0008}$. Relative to no alignment, constrained registration reduces nuisance penalties by 30.118 points while preserving 11.189 additional points of structural penalty. The frozen production score, coverage-aware audit, and ASDS obtain Spearman correlations of 0.297, 0.429, and 0.556 with mean ratings; grouped out-of-fold ASDS retains 0.540. Parsed-union and direct-ink ASDS are nearly equivalent ($\boldsymbol{\Delta\rho = 0.0004}$, 95% CI $\mathbf{-0.0101}$–$\mathbf{0.0103}$). The computer-vision contribution is a reproducible framework for overlapping dense parsing, unseen-class evaluation, discrepancy-preserving registration, and human-audited comparison under limited labelled data, rather than a claim of universal calligraphic beauty.

Keywords: Chinese calligraphy, document analysis, multi-label segmentation, reference-based assessment, human validation, explainable feedback

1 Introduction

Chinese calligraphy (*shufa*) is both a writing practice and an important form of Chinese traditional culture. Its visual identity is carried not only by the character that is written, but also by the organization of strokes, occupied and unoccupied space, balance, proportion, and relations between stroke endpoints. For readers without a Chinese-language background, a finished character can be understood as a structured composition of horizontal, vertical, rising-diagonal, falling-diagonal, and compound/curved stroke families. These components frequently adhere or cross in the final raster even though they arise from distinct writing actions. Consequently, two images can be recognized as the same character while differing substantially in the local structure that a learner needs to revise.

Most computer-vision systems for Chinese handwriting address recognition, style classification, or glyph generation. These outputs are useful but do not answer the assessment question: given a learner image and a designated same-character reference, which structural regions agree, which are missing or excessive, and which global differences should be tolerated? Recognition reduces the image to an identity; style classification reduces it to a class; and generation optimizes visual plausibility rather than verifying the learner's local discrepancies. A practice-oriented system instead needs spatial evidence that can be inspected before it is verbalized.

This requirement exposes four broader computer-vision difficulties. First, standard softmax segmentation assumes one semantic label per pixel. At a calligraphic crossing, however, one pixel can legitimately support multiple directional structures. Treating the channels as mutually exclusive destroys the annotation geometry, while unconstrained instance segmentation is poorly matched to a low-resource corpus without reliable ordered-stroke identities. Second, direct pixel overlap confounds structure with nuisance translation, global scale, and small rotation. Third, highly flexible registration solves the opposite problem too aggressively: anisotropic or deformable warps can align away a misplaced or disproportionate region that the assessment should retain. Fourth, a score that is stable under synthetic transformations is not automatically valid as a human-facing measure. It can reward empty channels, depend on the available character distribution, or disagree with perceived structural organization.

These are not calligraphy-specific inconveniences. They recur in document and shape analysis whenever layered or intersecting structures must be parsed from limited dense annotations, a query must be compared with a reference under nuisance geometry, and the final discrepancy must remain interpretable. Examples include touching handwriting, overlapping diagram primitives, line-art components, and other sparse foreground structures for which registration and diagnosis cannot be optimized independently. The central computer-vision problem is therefore to combine overlapping representation, low-resource generalization measurement, discrepancy-preserving registration, and human-audited comparison without using success in one stage as evidence for another.

The available data make this separation especially important. The recovered segmentation corpus contains complete six-channel ground truth for only 40 character identities, and reliable writer and style provenance cannot be reconstructed. A random or identity-overlapping split can therefore produce a strong in-domain score while leaving generalization to unseen character structures unresolved. We retain a quality-controlled standard protocol but also freeze a character-disjoint protocol, following the broader document analysis principle that performance under limited labels and source constraints should be measured directly rather than inferred [1, 16].

We study four research questions:

RQ1: How accurately do convolutional and transformer parsers recover overlapping direction and endpoint masks, and how much performance is lost when character identities are unseen during training?
RQ2: Does constrained global registration reduce nuisance sensitivity relative to no alignment while preserving penalties for local structural changes, and what semantic weaknesses remain in the deployed score?
RQ3: How strongly do fixed reference-conditioned scores associate with blinded human structural-similarity ratings, and what can be concluded from a retrospectively developed spatial-distribution score and its direct-ink ablation?
RQ4: Which direction and location evidence can be diagnosed reliably, and which observed failures arise from the model, the evidence representation, or the evaluation metric?

Our pipeline separates perception, comparison, and presentation:

$$\begin{aligned}\text{overlapping parsing} &\rightarrow \text{constrained registration}\\ &\rightarrow \text{structural comparison}\\ &\rightarrow \text{localized evidence}\\ &\rightarrow \text{text realization.}\end{aligned}$$

The parser produces five independently activated direction channels and one endpoint channel. The deployed comparison retains a frozen Dice/IoU/keypoint formula. We additionally examine aligned spatial-distribution similarity (ASDS), a compact silhouette score based on Jensen–Shannon similarity between polar occupancy, coarse grid occupancy, and horizontal/vertical projection profiles. Because ASDS was designed after the original human-rating study, its full-sample and character-grouped out-of-fold results are reported as retrospective development and internal validation, not as external confirmation.

The computer-vision contributions are:

1. an overlapping six-channel output contract, with independent activations at intersecting pixels, and a reproducible semantic QC protocol for low-resource complete-character parsing;
2. a three-architecture, three-seed comparison on matched standard and character-disjoint splits, exposing architecture-dependent differences between dense direction parsing and sparse endpoint localization;
3. a restricted reference-registration policy evaluated with controlled nuisance and structural perturbations, paired uncertainty, and a wider-search negative ablation, thereby measuring reduction rather than claiming transform invariance;
4. an auditable assessment analysis comprising inactive-channel semantics, cross-reference tests, a blinded 150-pair human-rating cohort, and transparent retrospective ASDS development with grouped internal validation;
5. a paired direct-ink ablation that separates silhouette-only scoring from the parser's independent role in direction-, overlap-, and endpoint-level evidence, plus a diagnostic failure taxonomy that keeps evidence correctness separate from language realization.

The intended input is a raster exported from a direct digital writing canvas, not a photograph of paper. The claims therefore concern structural analysis in this modality. They do not cover camera robustness, writer-disjoint generalization, stroke-order recovery, historical authenticity, or universal aesthetic grading.

2 Related Work

2.1 Recognition, document segmentation, and stroke parsing

Large databases such as CASIA-HWDB established modern benchmarks for online and offline handwritten Chinese character recognition [13]. Recognition assigns a character identity, whereas our task requires spatially resolved evidence about directional stroke regions and keypoints. It is therefore closer to document segmentation than to closed-set classification.

U-Net introduced an encoder–decoder architecture with skip connections [18]; DeepLabV3+ combines atrous spatial pyramids with a decoder [3]; and SegFormer uses hierarchical transformer features with a lightweight multilayer-perceptron decoder [23]. Generic document models such as dhSegment illustrate the value of reusable segmentation formulations [16]. Recent IJDAR work further shows that label scarcity, source-target mismatch, and the choice of pretraining protocol can materially affect handwritten document segmentation [1]. We use established architectures to study a different output contract: the five direction labels are independently activated because intersections can support more than one label.

Stroke-Seg is the most directly related recent framework for Chinese character stroke segmentation [8]. It combines synthetic-image generation with interchangeable semantic-segmentation backbones to separate character strokes. Our task differs in annotation geometry and downstream contract. Each manually isolated stroke is assigned by a fixed character-specific mapping to one of five direction families; crossings may therefore activate multiple channels at the same pixel, and a sixth mask records annotated stroke endpoints. We do not infer an ordered stroke list. Instead, the representation preserves overlapping structural evidence for reference-conditioned comparison. A visually plausible foreground partition can otherwise violate the multi-label geometry of a crossing or lose endpoint evidence required for diagnosis.

Computational calligraphy research spans handcrafted style descriptors, recognition, generation, and feature extraction. Earlier work represented calligraphic style using local and global shape features [17, 26]. CalliNet advances style classification with metric learning and triplet supervision [25]. Other systems extract artistic regions from a calligraphy work relative to a reference [19], beautify handwritten glyphs toward a Kai-style template [22], or evaluate robotic writing through designed aesthetic features [14].

Large contemporary resources broaden the available calligrapher, script, and page-level annotations. UniCalli, for example, supports column-level calligraphy generation and related representation learning [24]. These resources are valuable for style generation and recognition, but they do not remove the need to validate a practice score against human judgments. Our present study is deliberately narrower: it compares two instances of the same character against a selected reference and reports structural evidence rather than claiming author authentication or style generation.

2.2 Reference-conditioned assessment and shape comparison

A recent survey divides handwritten Chinese character quality evaluation into rule-based geometric measures, template comparison, and learned methods, while emphasizing the difficulty of defining quality labels and explaining model decisions [4]. Pairwise and Siamese approaches can learn similarity from labelled examples [5]. Multidimensional calligraphy evaluation has also been studied using visual, quality, and aesthetic attributes [20]. These approaches demonstrate that "quality" is not a single observable quantity: legibility, geometric correctness, reference similarity, brushwork, and artistic appeal can disagree.

Our score is consequently scoped as *reference-conditioned structural agreement*. The human study asks raters to judge same-character structural similarity, not beauty or historical fidelity. This distinction is central to the validity claim: a moderate association with structural ratings cannot be relabelled as expert aesthetic calibration.

Registration methods range from parametric geometric alignment to learned spatial transformers [9, 27]. Flexible registration is useful when deformation is a nuisance, but it is hazardous for diagnostic assessment: anisotropic or deformable warping can conceal a proportion or placement error. We therefore restrict alignment to translation, isotropic scale, and small rotation and evaluate that policy with both nuisance and structural perturbations.

Shape contexts describe the spatial distribution of contour points in log-polar bins [2]; spatial pyramids retain coarse layout that an orderless representation would discard [11]. ASDS follows the same general principle but uses binary-ink occupancy rather than point correspondences. Its distributions are compared with the bounded, symmetric Jensen–Shannon divergence [12]. Polar occupancy captures relative radial/angular organization, a 3×3 grid retains coarse canvas layout, and projection profiles retain row/column mass. The design is simpler than a learned quality regressor and remains directly inspectable. Alignment and comparison are evaluated separately: useful registration reduces nuisance sensitivity without erasing controlled structural discrepancies, rather than merely maximizing overlap.

2.3 Human validation and diagnostic evidence

Automated assessment requires both association with human judgment and a clear account of rater reliability. We report two-way random-effects, absolute-agreement intraclass correlations following established ICC definitions and reporting guidance [10, 15]. Uncertainty for score correlations is estimated by bootstrap resampling at the character level [7]; paired alignment comparisons additionally use the Wilcoxon signed-rank test [21].

Explanation is evaluated separately from score validity. The system first emits normalized evidence—direction channel, missing or extra structure, spatial region, centre offset, scale ratio, and endpoint agreement. A language model may verbalize this evidence, but it does not determine the score. This separation prevents fluent text from being treated as evidence of diagnostic correctness and reflects a general requirement for human-facing computer-vision systems: semantic attribution and localization should be validated before natural-language fluency.

3 Method

3.1 Problem formulation

Let I_u be a learner image and I_r an approved reference image of the same target character. The system first predicts overlapping structural masks, then aligns the reference to the learner with a restricted global transform. It returns (i) six masks, (ii) a scalar structural-agreement score, and (iii) localized evidence (Fig. 1). Cross-character scoring is prohibited by the application contract.

3.2 Overlapping six-channel parsing

For an RGB character image I, a parser f_θ predicts

$$P = f_\theta(I) \in [0,1]^{H\times W\times 6}.$$

The fixed channel order is

$$(\text{vec1}, \text{vec2}, \text{vec3}, \text{vec4}, \text{vec5}, \text{keypoint}).$$

The first five channels have the following fixed meanings: vec1 is vertical, vec2 is a rising diagonal (/), vec3 is horizontal, vec4 is a falling diagonal (\), and vec5 contains compound, curved, or otherwise non-cardinal strokes that cannot be assigned to the first four families. The sixth channel is the union of the two manually marked endpoints from each isolated stroke image. It does not encode junctions, turning points, or stroke order.

The ground-truth protocol begins with the completed character and its individually isolated stroke images. A fixed, character-specific `STROKE_VECTOR_MAP` assigns each isolated stroke to one of the five direction families; pixels from strokes assigned to the same family are merged. The endpoint annotations from all isolated strokes are merged into the keypoint mask. Because independently acquired strokes may occupy the same pixel after composition, a crossing can legitimately activate multiple direction channels. Each channel therefore uses an independent sigmoid rather than a mutually exclusive softmax (Fig. 2).

Images are converted to RGB, letterboxed to 512×512, and normalized. Masks are resized by nearest-neighbour interpolation. For direction channel c, the loss combines class-weighted binary cross entropy and soft Dice:

$$\mathcal{L}_{\text{dir}} = \sum_{c=1}^{5} \left(\mathcal{L}_{\text{BCE}}^{(c)} + \mathcal{L}_{\text{Dice}}^{(c)}\right).$$

The sparse keypoint target uses focal loss with $\gamma = 2$ and $\alpha = 0.75$, plus Dice loss. A boundary term derived from the masks is weighted by 0.2:

$$\mathcal{L} = \mathcal{L}_{\text{dir}} + \mathcal{L}_{\text{focal,kp}} + \mathcal{L}_{\text{Dice,kp}} + 0.2\mathcal{L}_{\text{boundary}}.$$

3.3 Reference cache and constrained registration

Approved reference images are parsed once and stored as binary $512 \times 512 \times 6$ arrays. Each cache entry records its reference ID, character, course/style category, parser version, checkpoint SHA-256, thresholds, and channel schema. Caching removes repeated GPU inference for the reference branch. These masks are model-derived references and are never treated as segmentation ground truth.

Constrained similarity registration. Let U and R denote the learner and reference binary stacks, and let

$$M(X) = \bigvee_{c=1}^{5} X_c$$

be the union of direction ink. For each candidate isotropic scale s and rotation ϕ, the transformed reference is translated so that its foreground centroid approaches the learner centroid. The production search is bounded by

$$s \in [0.8, 1.2], \qquad \phi \in [-3^\circ, 3^\circ],$$

and selects the candidate maximizing

$$J\big(M(U), M(T_{s,\phi,t}(R))\big),$$

where J is foreground IoU. Nonuniform scale, shear, thin-plate splines, and deformable warping are disallowed. This bounded search is a conservative compensation policy: it can reduce penalties from page placement and modest global size or rotation differences, while leaving local proportion and stroke-region errors available for diagnosis. It

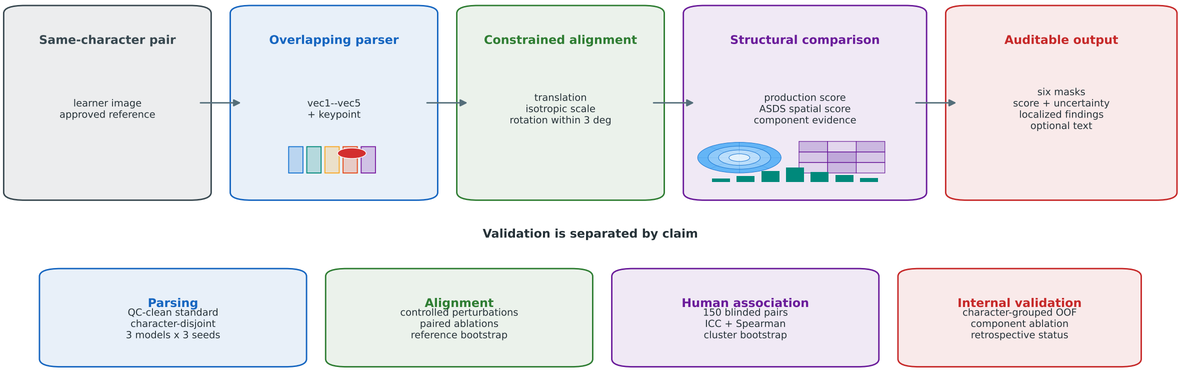

Fig. 1 System and validation overview. The reference and learner are parsed into overlapping direction/keypoint masks, globally aligned under a restricted transform, and compared by the frozen production score and ASDS. Segmentation, alignment, and human association are evaluated by separate protocols so that one successful experiment is not used to support a different claim.

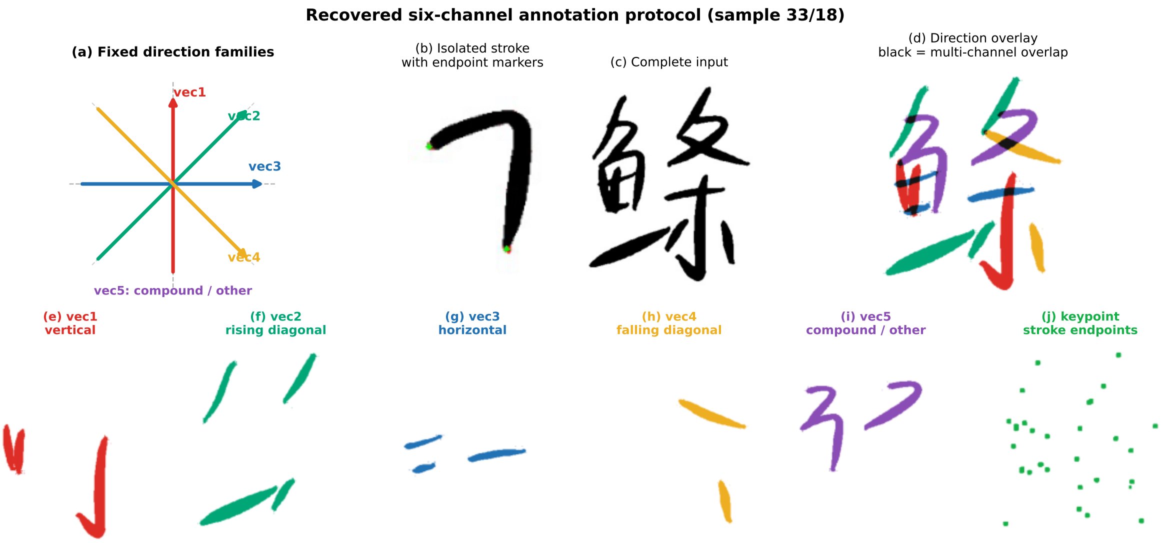

Fig. 2 Ground-truth six-channel definition. The direction guide defines vertical (vec1), rising-diagonal (vec2), horizontal (vec3), falling-diagonal (vec4), and compound/curved/other (vec5) families. Isolated stroke images are assigned through a fixed character-specific mapping and composed into overlapping masks; black in the colour overlay denotes pixels shared by more than one direction channel. The keypoint mask is the union of annotated stroke endpoints only. Endpoint markers are dilated in this figure for print visibility; the stored ground truth is unchanged.

does not imply invariance to all transforms within the nominal range.

3.4 Structural agreement scores

Frozen production score.

After alignment, let D_c be Dice agreement in direction channel c,

$$\overline{D} = \frac{1}{5}\sum_{c=1}^{5} D_c,$$

let J_{ink} be union-ink IoU, and let $F_{\text{kp}}^{(3)}$ be keypoint-mask F1 with a three-pixel tolerance.

The deployed score is

$$S_{\text{prod}} = 100\left(0.55\overline{D} + 0.25J_{\text{ink}} + 0.20F_{\text{kp}}^{(3)}\right).$$

This formula was frozen before the controlled experiments. An audit variant excludes direction channels that are empty in both images and avoids granting perfect keypoint credit when both keypoint masks are empty. The audit score is not retroactively substituted for S_{prod}.

Aligned spatial-distribution similarity. The human study showed that pixel- and channel-overlap terms incompletely capture perceived global organization. We therefore develop aligned spatial-distribution similarity (ASDS) on the union direction ink after the same frozen registration. The score uses three complementary distributions.

Polar occupancy. For a binary ink mask M, foreground coordinates are centred at their ink centroid. Radius is divided by the 95th percentile of foreground radius and clipped to $[0, 1)$. Each pixel is assigned to one of four radial and eight angular bins, producing a normalized 32-bin histogram $h_p(M)$. Independent centroid and radius normalization make this term resistant to small residual translation and isotropic scale, while retaining angular and radial mass organization.

Coarse grid occupancy. The canonical canvas is partitioned into a 3×3 grid. Normalized foreground counts form $h_g(M)$. Unlike the centroid-normalized polar term, this component retains coarse location and balance on the writing canvas.

Projection profiles. Normalized row sums $h_y(M)$ and column sums $h_x(M)$ describe vertical and horizontal mass allocation. Their similarities are averaged.

For distributions p and q, with $m = (p + q)/2$, we use

$$D_{\text{JS}}(p, q) = \frac{1}{2}D_{\text{KL}}(p\|m) + \frac{1}{2}D_{\text{KL}}(q\|m),$$

with base-2 logarithms, and define

$$\text{JSS}(p, q) = 1 - \sqrt{D_{\text{JS}}(p, q)}.$$

Because $D_{\text{JS}} \in [0, 1]$, $\text{JSS} \in [0, 1]$. The component similarities are

$$s_p = \text{JSS}(h_p(U), h_p(R)),$$
$$s_g = \text{JSS}(h_g(U), h_g(R)),$$

$$s_{\text{proj}} = \frac{1}{2}\big[\,\text{JSS}(h_x(U), h_x(R)) + \text{JSS}(h_y(U), h_y(R))\big].$$

The frozen development specification is

$$S_{\text{ASDS}} = 100\big(0.70s_p + 0.15s_g + 0.15s_{\text{proj}}\big).$$

The development search placed 0.70 weight on the polar term. Rather than retain the more sample-specific unequal secondary weights, we split the remaining 0.30 equally between grid and projection evidence before freezing the reported specification.

The score does not use character ID, writer ID, style ID, or a human rating at inference. Nevertheless, its feature choice and weights were finalized after observing the original 150-pair study. It is therefore a development-stage algorithm with resampling-based internal validation rather than an independently confirmed score. Features, bins, weights, alignment, candidate search, and selection policy are reported so the remaining risk of retrospective optimism is explicit.

ASDS uses only the union of the five direction masks. Consequently, a six-channel parser is not mathematically necessary merely to construct the binary silhouette consumed by ASDS in the present direct-digital modality. ASDS is one scalar consumer of the parsing pipeline, rather than the sole justification for it. The separate channels remain necessary for the deployed production score, per-direction agreement, missing/extra directional evidence, endpoint relations, and localized feedback. We therefore include a paired direct-ink ASDS ablation that applies the same alignment, descriptors, and frozen weights directly to thresholded source ink. This ablation isolates whether parsing supplies any measurable normalization benefit to the scalar score; it does not test the channel-level outputs that require parsing.

3.5 Evidence extraction and feedback

The system preserves component scores, selected transform, pre-alignment centre distance, ink-area ratio, per-direction agreement, and keypoint agreement. Difference masks are summarized by direction channel, missing reference ink versus extra learner ink, and a 3×3 region set. Connected components of the five direction masks are exposed as direction-classified regions, not as recovered stroke order.

Deterministic feedback ranks centre, scale, keypoint, and local-direction findings. An optional text model receives only this structured contract and may improve fluency; it cannot change the score or invent unsupported brushwork or stroke-order claims.

4 Experimental Protocol

4.1 Data resources and evaluation cohorts

Project-collected segmentation corpus. The original collection protocol commissioned contributors with calligraphy training to write complete characters and their isolated component strokes on a direct digital canvas. The isolated-stroke images were assigned through the fixed character-specific mapping described in Sect. 3, and their manually marked endpoints were merged to produce six-channel ground truth. The labels are therefore derived from captured stroke components, not from predictions by the models evaluated in this paper. The historical records do not preserve a trustworthy mapping from samples to individual writers or a standardized measure of contributor expertise; we consequently report the commissioned, calligraphy-trained collection procedure but do not claim professional-calligrapher, different-writer, or writer-disjoint strata.

Recovery yielded 894 sample directories across 43 character IDs. Complete RGB images, five direction masks, an endpoint mask, and a stacked six-channel mask are available for 840 samples from character IDs 0–39. Fifty-four samples from IDs 40–42 lack the character-specific direction mapping and are excluded rather than pseudo-labelled. A frozen semantic QC procedure further identified 12 source-image/GT mismatches and 59 non-canonical members of exact image-and-mask duplicate groups. These exclusion sets do not overlap, leaving 769 unique QC-clean observations. Duplicate membership and mismatch decisions were fixed before the formal model comparison, and no exact duplicate group crosses the original train/validation/test assignment.

The standard assignment contains 600/120/120 samples before QC and 530/119/120 after QC. A character-disjoint assignment was frozen before formal evaluation with 28/6/6 disjoint character identities and 588/126/126 samples; the same QC exclusions leave 539/114/116 train/validation/test observations. Source assignments, derived splits, and exclusion contracts are content-hashed, with full SHA-256 values in the reproducibility supplement. Because writer identity is unavailable, neither split supports a writer-disjoint claim.

Reference and perturbation resources. The reference-conditioned experiments use 200 approved images from the Calli-Tongji Beta release: 100 in the Ouyang Xun regular-script category and 100 in the Wang Xizhi running-script category. The frozen SegFormer-B2 v1 parser produces a versioned six-mask cache for these references. Each entry records its source ID, character, category, thresholds, parser version, and checkpoint hash. These cached predictions are reference representations, not segmentation ground truth. The controlled-perturbation and registration cohorts are generated deterministically from the same 200 cached real-reference masks, allowing score behaviour to be studied without confounding it with a second parser inference.

Cross-reference and diagnostic cohorts. The reference library supports seven natural same-character cross-style pairs. All seven are retained as the positive cross-reference set; the library contains no same-style, different-instance reference pair. Fifty different-character negatives are selected independently of model score. The diagnostic cohort is generated from prespecified deletion, addition, shift, dilation, erosion, and endpoint perturbations. Its ground truth is the known intervention rather than a human pedagogical label.

Human-rating development cohort. Before ratings were collected, 150 natural same-character, different-instance pairs were selected from the 769 QC-clean project samples. All 40 character identities are represented by three or four pairs. Source/GT mismatches, exact duplicates, suspected near duplicates, and repeated reference/candidate instances were excluded. Pair roles, asset hashes, blinded identifiers, and display order were frozen, and all automated scores and source identifiers were hidden from raters.

Three consenting adult raters independently judged structural similarity on a five-point ordinal scale. Their self-reported backgrounds were heterogeneous: one reported one year of calligraphy learning/teaching experience, one reported two years of other relevant experience, and one reported no relevant experience. We therefore use "human raters", not "calligraphy experts". Each rater evaluated 150 canonical pairs plus 15 randomly interspersed hidden repeats. This cohort is the development and internal-validation resource for ASDS; it is not an independent external test.

4.2 Parsing training and evaluation

We compare a 32-base-channel U-Net, ResNet-50 DeepLabV3+, and ImageNet-pretrained SegFormer-B2. Each architecture is run with seeds 20260811, 314159, and 271828 on both split protocols, for 18 formal runs. Inputs are 512×512. Batch size is 8 for U-Net and 4 for the two larger models. Label-safe augmentation includes translation up to 12 pixels, isotropic scale 0.9–1.1, brightness 0.85–1.15, contrast 0.9–1.1, and Gaussian blur with probability 0.15.

All models use AdamW, weight decay 0.01, automatic mixed precision, gradient clipping at 1.0, three warm-up epochs, cosine decay to 10^{-6}, at most 120 epochs, and early stopping with patience 15. U-Net uses learning rate 3×10^{-4}. DeepLabV3+ uses encoder/decoder rates $10^{-4}/10^{-3}$; SegFormer-B2 uses $3 \times 10^{-5}/3 \times 10^{-4}$. Positive BCE weights are estimated from at most 200 training samples and capped at 100. Thresholds are calibrated on the validation split only.

We report five-direction Macro Dice and Macro IoU, per-channel precision/recall/Dice, Boundary F1, strict endpoint F1, and tolerant endpoint F1 at one, three, and five pixels. Test results are summarized by the mean and sample standard deviation across all declared seeds. The standard protocol measures in-corpus instance generalization, whereas the character-disjoint protocol measures transfer to held-out character identities.

4.3 Reference-conditioned robustness and diagnostic evaluation

Controlled perturbations are applied directly to cached reference masks. Nuisance families comprise translation, rotation, scale up, scale down, and a compound allowed transform. Structural families comprise terminal deletion, added fragment, local fragment shift, direction dilation, direction erosion, and endpoint shift. Target channels and movement axes are selected by stable hashes rather than model error or manual case choice. A transformed case that would risk canvas clipping is marked invalid with a recorded reason and is excluded from valid-score summaries. For every family we report valid N, mean, median, sample standard deviation, 5th and 95th percentiles, 95% reference-bootstrap confidence intervals, invalid fraction, within-reference severity Spearman correlation, and adjacent monotonicity.

The identical reference/perturbation/severity tuples are evaluated with no alignment, the production constrained alignment, and a preregistered wider search with $s \in [0.6, 1.4]$ and $\phi \in [-12^\circ, 12^\circ]$. Comparisons are paired. Confidence intervals resample references; Wilcoxon signed-rank tests and rank-biserial effect sizes use per-reference mean paired differences. Family-level comparisons are the primary ablation, while perturbation-level p values are unadjusted descriptive results.

The score audit counts inactive direction channels and recomputes the coverage-aware candidate in Sect. 3.4. Qualitative audit cases are selected by correction quantiles rather than appearance. Cross-reference semantics are measured on all seven available same-character cross-style pairs and the 50 score-independent different-character negatives; self-pairs are prohibited and the unavailable same-style/different-instance condition is not synthesized.

For diagnostic evaluation, the perturbation cause supplies deterministic ground truth. We report Recall@3, Top-1 cause, direction channel, missing/extra type, exact and overlapping grid localization, specificity, and centre-direction wording. Language-model realization is disabled. Because the benchmark assesses engineered interventions rather than expert pedagogy, it is treated as a secondary diagnostic analysis.

4.4 Human-audited score validation and retrospective ASDS development

The frozen production and coverage-aware scores are compared with the three-rater mean on the 150 canonical pairs using Spearman correlation. Confidence intervals use 10,000 bootstrap resamples clustered by target character. Inter-rater reliability uses two-way random-effects, absolute-agreement ICC(2,1) and ICC(2,k). Hidden repeats are excluded from correlation and ICC and used only for exact agreement, agreement within one point, mean absolute difference, and quadratic weighted kappa.

Post-rating ASDS development. The original ratings form the ASDS development set. Three fixed distribution families are considered: polar occupancy, 3×3 grid occupancy, and projection profiles. A coarse non-negative simplex search evaluates component weights. The search-selected polar weight is retained, while the remaining weight is split equally between grid and projection terms to reduce sample-specific tuning. The full-grid apparent optimum is $(w_g, w_p, w_{\text{proj}}) = (0.225, 0.700, 0.075)$ with $\rho = 0.557$; the frozen 0.15/0.70/0.15 specification gives $\rho = 0.556$. Five-fold internal validation groups by character identity, and weights for each fold are selected using the other characters only. Character-cluster bootstrap intervals are reported for the frozen full-sample formula, grouped out-of-fold predictions, and paired differences from the production and coverage-aware scores.

Polar, grid, and projection components are also reported separately, and the combined score is compared with each component. These development comparisons are exploratory and are not multiplicity-adjusted. The complete candidate search is retained so that post-rating researcher degrees of freedom remain visible.

Paired direct-ink ASDS ablation. To test whether the scalar ASDS result itself requires six-channel parsing, we recompute it on the same 150 pairs using foreground extracted directly from the original direct-digital RGB rasters. Grayscale pixels with intensity below 240—the threshold already frozen for dataset QC—define foreground; no learned preprocessing is used. Direct-ink masks pass through the same constrained alignment and the same 0.70/0.15/0.15 formula. Threshold, alignment, features, and weights are not selected against the ratings for this comparison.

We report Spearman association for parsed-union and direct-ink ASDS, their paired difference, and their mutual rank association, with 10,000 paired bootstrap resamples clustered by character identity. Because this ablation uses the original development pairs, it measures implementation dependence within the same retrospective cohort rather than external validity.

5 Results

5.1 RQ1: segmentation and unseen-character generalization

Table 1 shows that, on the QC-clean standard split, DeepLabV3+ obtains the highest direction Macro Dice (0.9630 ± 0.0006), Macro IoU (0.9287 ± 0.0012), and Boundary F1 (0.8443 ± 0.0028). SegFormer-B2 is close in direction overlap but shows greater between-seed variation. U-Net is weaker on direction and boundary metrics yet gives the best strict keypoint F1 (0.7562 ± 0.0038). Thus, dense direction segmentation and sparse keypoint localization favour different architectures.

All models degrade when character identities are held out (Table 2). SegFormer-B2 gives the strongest character-disjoint direction result: Macro Dice 0.8866 ± 0.0008, Macro IoU 0.7977 ± 0.0013, and Boundary F1 0.6851 ± 0.0046. Its Macro-Dice decrease from the standard split is 7.150 percentage points, versus 9.664 for DeepLabV3+ and 9.360 for U-Net. DeepLabV3+ remains second for direction overlap, while U-Net retains the highest strict keypoint F1. SegFormer-B2 is therefore the strongest evaluated model for unseen-character direction parsing, not a universal winner across all six outputs.

The earlier single SegFormer release achieved direction Macro Dice 0.9610 but was trained before semantic QC. No exact duplicate group crosses its standard split, so its result is not

Table 1 QC-clean standard-split segmentation results over three fixed random seeds. Values are mean ± sample standard deviation.

Model	Macro Dice	Macro IoU	Keypoint F1	Boundary F1
U-Net	0.9196 ± 0.0018	0.8515 ± 0.0030	$\mathbf{0.7562 \pm 0.0038}$	0.8009 ± 0.0023
DeepLabV3+	$\mathbf{0.9630 \pm 0.0006}$	$\mathbf{0.9287 \pm 0.0012}$	0.7218 ± 0.0012	$\mathbf{0.8443 \pm 0.0028}$
SegFormer-B2	0.9581 ± 0.0031	0.9196 ± 0.0057	0.7139 ± 0.0046	0.8259 ± 0.0088

Table 2 Character-disjoint test results over three fixed random seeds. Train, validation, and test character identities are disjoint. Values are mean ± sample standard deviation.

Model	Macro Dice	Macro IoU	Keypoint F1	Boundary F1
U-Net	0.8260 ± 0.0161	0.7120 ± 0.0225	$\mathbf{0.6972 \pm 0.0045}$	0.6430 ± 0.0136
DeepLabV3+	0.8664 ± 0.0094	0.7663 ± 0.0146	0.6755 ± 0.0031	0.6728 ± 0.0138
SegFormer-B2	$\mathbf{0.8866 \pm 0.0008}$	$\mathbf{0.7977 \pm 0.0013}$	0.6604 ± 0.0033	$\mathbf{0.6851 \pm 0.0046}$

explained by exact train–test duplication. Nevertheless, its training population included 12 mismatched and duplicate-weighted observations; all formal comparisons above use the frozen 769-sample QC-clean population.

5.2 RQ2: registration, score behaviour, and semantic audits

The family-level results in Table 3 show heterogeneous residual nuisance sensitivity. Translation is recovered exactly in all 800 valid observations. Mean absolute penalties are 5.395 for rotation, 13.653 for scale-up, and 16.356 for scale-down. Scale-up and compound transformations have invalid fractions of 49.0% and 54.0%, respectively, because transformed support risks leaving the canvas; those cases are reported rather than silently counted as score failures. Scale-related summaries are therefore conditional on clipping-safe references. Because clipping safety is related to glyph extent, these conditional means should not be interpreted as full-library invariance.

Structural perturbations generally lower the score with severity. Direction dilation and erosion have the largest average penalties, followed by terminal deletion and added fragments. Local fragment shifts and keypoint shifts are penalized more weakly. The benchmark therefore identifies both a desired signal—sensitivity to local structure—and a principal weakness—residual scale sensitivity. Full per-family distributional statistics are provided in Supplementary Table S1.

The paired comparison in Table 4 shows that, relative to no alignment, the constrained method reduces nuisance penalties by 30.118 points (reference-bootstrap 95% CI 29.506–30.727) and preserves an additional 11.189 points of structural penalty (10.094–12.322). Both per-reference signed-rank comparisons are nonzero with rank-biserial effect size 1.0.

The wider alignment does not dominate the production range. The current range has a 2.896-point lower nuisance penalty (2.757–3.034), whereas the wider range preserves 0.284 additional points of structural penalty. A wider search therefore fails to reduce residual nuisance sensitivity and provides only a small structural-sensitivity advantage. The paired family-level breakdown is reported in Supplementary Table S2.

Score semantics and cross-reference tests. Table 5 shows that 41 of 200 references contain an inactive direction channel: 36 contain one and five contain two; none contains three or more. Empty-channel agreement affects 1,252 valid perturbation rows. Replacing the all-five mean by an active-channel mean changes the score by -0.289 points on average and by as much as -15.402 points. The largest corrections occur when inactive channels receive perfect Dice credit while a perturbed active channel degrades.

Figure 3 illustrates the prespecified correction quantiles without selecting examples for visual appeal.

The seven same-character cross-style pairs in Table 6 have mean score 20.239 (bootstrap 95% CI 15.639–25.069), compared with 9.741 (8.584–10.970) for 50 different-character negatives. Cliff's

Table 3 Controlled perturbation summary. Nuisance rows report absolute score drop; structural rows report signed drop from identity. Full distribution and severity statistics are in Supplementary Table S1.

Perturbation	Valid N	Mean drop	95% CI	Invalid fraction	Monotonic
Translation	800	0.000	[0.000, 0.000]	0.000	1.000
Rotation	520	5.395	[5.079, 5.747]	0.133	0.647
Scale up	408	13.653	[12.964, 14.420]	0.490	0.271
Scale down	800	16.356	[15.670, 17.074]	0.000	0.500
Compound allowed transform	276	8.143	[7.621, 8.717]	0.540	0.466
Terminal deletion	800	19.757	[17.387, 22.087]	0.000	0.992
Added direction fragment	800	11.398	[9.934, 12.929]	0.000	0.987
Local fragment shift	800	4.310	[3.886, 4.772]	0.000	0.985
Direction dilation	800	24.034	[22.180, 25.955]	0.000	1.000
Direction erosion	800	22.225	[20.269, 24.344]	0.000	0.997
Keypoint shift	800	7.468	[7.294, 7.641]	0.000	1.000

Table 4 Paired alignment comparison. Positive benefit differences favour the current constrained alignment. Wilcoxon tests use per-reference mean paired differences.

Comparison	Family	Refs	Ours	Comparator	Benefit 95% CI	p	RBC
Ours–none	nuisance	200	8.455	38.573	[29.506, 30.727]	< 0.001	1.000
Ours–none	structural	200	14.865	3.677	[10.094, 12.322]	< 0.001	1.000
Ours–wide	nuisance	200	8.455	11.351	[2.757, 3.034]	< 0.001	1.000
Ours–wide	structural	200	14.865	15.149	[-0.355, -0.213]	< 0.001	-0.683

Table 5 Inactive direction channels among 200 references.

Inactive channels	References	Fraction
0	159	0.795
1	36	0.180
2	5	0.025

delta is 0.84 [6]. This is evidence of a same-character reference signal, but the small positive set and missing same-style/different-instance condition prevent a broad style-generalization claim.

5.3 RQ3: blinded human validity and ASDS development

Across 150 canonical pairs, S_{prod} has a positive but modest association with the three-rater mean ($\rho = 0.297$, character-cluster bootstrap 95% CI 0.148–0.441, $p = 0.00023$). The coverage-aware audit is stronger ($\rho = 0.429$, 0.313–0.535, $p < 10^{-7}$). The paired bootstrap difference is 0.132 (0.062–0.197). This result is consistent with the inactive-channel audit but does not, by itself, authorize post-hoc replacement of the production score.

These associations are summarized in Table 7. Canonical ICC(2,1) is 0.448 and ICC(2,k) is 0.709 (Table 8). Across 45 rater-level hidden repeats, exact agreement is 48.9%, agreement within one scale point is 93.3%, mean absolute difference is 0.578, and pooled quadratic weighted kappa is 0.634. Individual production-score correlations are positive for all raters and range from 0.174 to 0.350. Average ratings are therefore more reliable than a single rating, but meaningful subjectivity remains.

ASDS raises the development-set association to $\rho = 0.556$ (character-cluster bootstrap 95% CI 0.454–0.651). Predictions produced by character-grouped five-fold weight selection retain $\rho = 0.540$ (0.432–0.639). The paired bootstrap interval for ASDS minus production is 0.110–0.410, and that for ASDS minus the coverage-aware audit is 0.017–0.239. For grouped out-of-fold predictions, the difference from coverage-aware scoring is less certain (-0.003–0.224).

The component analysis in Table 9 identifies the polar term as the strongest individual spatial descriptor ($\rho = 0.547$); grid and projection similarities reach 0.447 and 0.445. Thus, most ASDS association comes from radial/angular ink organization, while the two coarse canvas terms provide

Table 6 Cross-reference structural agreement.

Pair type	N	Mean	Median	Mean 95% CI
Same character, cross style	7	20.239	20.272	[15.639, 25.069]
Different-character negative	50	9.741	8.869	[8.584, 10.970]

Fig. 3 Coverage-correction cases selected by correction quantiles rather than visual preference. Purple denotes aligned overlap, blue missing-reference ink, and red extra-user ink.

Table 7 Association between the structural scores and the mean rating of three blinded human raters across 150 canonical pairs. Confidence intervals use 10,000 bootstrap resamples clustered by character.

Score	ρ	95% CI	p
Production	0.297	[0.148, 0.441]	2.28×10^{-4}
Coverage-aware	0.429	[0.313, 0.535]	4.32×10^{-8}

a small complementary contribution. The combined score is not claimed to be significantly better than the polar term alone.

These results show that human ratings are more closely related to spatial ink organization than to the deployed channel/pixel overlap alone. They are not an external validation of ASDS, because the same 150 ratings informed its feature design. The grouped out-of-fold estimate reduces weight-selection optimism and is reported as internal validation; the full-sample value remains descriptive and retrospective.

Table 8 Human-rater reliability. ICC uses only the 150 canonical presentations. Repeat statistics use the 15 hidden repeats per rater.

Rater	Exact	Within ±1	MAD	QWK
E01	0.600	1.000	0.400	0.559
E02	0.467	1.000	0.533	0.721
E03	0.400	0.800	0.800	0.591
POOLED	0.489	0.933	0.578	0.634

Canonical ICC(2,1)=0.448 and ICC(2,k)=0.709 for 150 pairs and 3 raters.

The paired direct-ink ablation (Table 10) gives $\rho = 0.55623$ for parsed-union ASDS and $\rho = 0.55583$ for direct-ink ASDS. Their difference is 0.00040, with character-cluster bootstrap 95% CI −0.01011–0.01030. The two score vectors are themselves strongly rank-associated ($\rho = 0.9973$). Thus, on these clean direct-digital inputs, the parser neither measurably improves nor degrades the scalar ASDS association. This result narrows the claim: ASDS is a silhouette score, whereas the parser is justified by the separate direction, overlap, and endpoint evidence evaluated in RQ1 and returned by the application.

Figure 4 visualizes the stronger monotonic organization of ASDS while preserving the substantial dispersion that motivates replication on independently collected writers and characters.

5.4 RQ4: diagnostic feedback

Table 11 separates the dominant localization failure types. Required Recall@3 is 0.701, whereas exact grid-region localization is 0.109. Among exact-region failures, 53.9% occur because the perturbation truth spans multiple cells while the metric requires one exact cell, 24.6% involve an incorrect direction channel, and 20.1% contain

Table 9 Development-stage ASDS component ablation and association with mean human structural rating. Confidence intervals use 10,000 character-cluster bootstrap resamples. ASDS was developed after the original ratings and is not an independent result.

Score or component	Spearman ρ	95% CI
Grid JS similarity only	0.447	[0.303, 0.572]
Projection JS similarity only	0.445	[0.316, 0.564]
Polar JS similarity only	0.547	[0.443, 0.645]
ASDS frozen development formula	0.556	[0.454, 0.651]
ASDS character-grouped OOF	0.540	[0.432, 0.639]

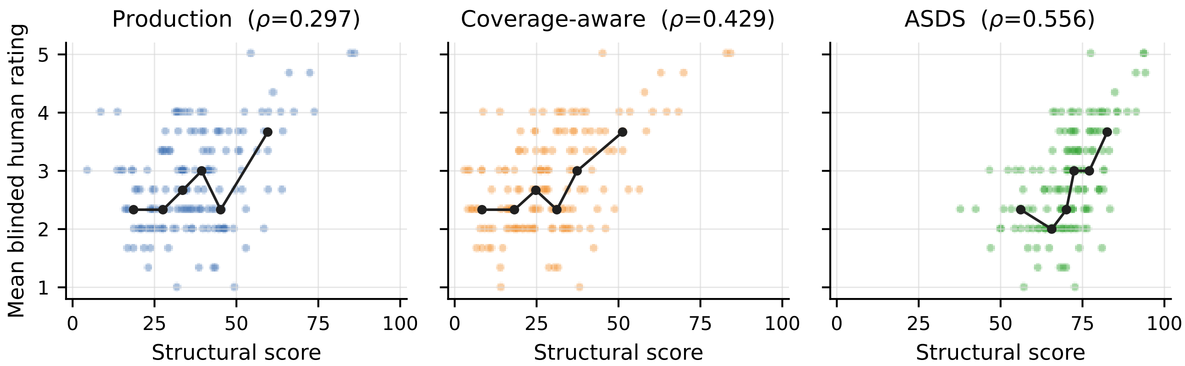

Fig. 4 Development-stage association between blinded mean structural rating and the production, coverage-aware, and ASDS scores for 150 natural same-character pairs. Points are individual pairs; lines connect score-bin medians only as a descriptive guide. ASDS was designed after these ratings, so this figure is not an independent validation.

Table 10 Direct-ink ablation of the frozen ASDS specification on the same 150 development pairs.

ASDS input	Spearman ρ	95% CI
Parsed union	0.556	[0.454, 0.651]
Direct ink	0.556	[0.454, 0.649]
$\Delta\rho$ (parsed–direct)	0.000	[-0.010, 0.010]

Both variants use the same alignment policy, ASDS features, and frozen weights. Direct ink uses the frozen dataset-QC rule: grayscale intensity < 240. No threshold or weight was selected against the ratings.

Table 11 Primary taxonomy of exact-region diagnostic failures.

Failure type	Count	Fraction
Multi-region/boundary truth	1922	0.539
Wrong direction channel	876	0.246
Local finding absent from Top-3	718	0.201
Wrong missing/extra type	38	0.011
Alignment residual shift	11	0.003

no local finding in the top three. Wrong missing/extra type and residual registration shifts are uncommon. The main limitation is therefore a mixture of localization representation and direction attribution rather than wording. This motivates a future set- or pixel-overlap localization metric, not further tuning of feedback rules on the same cases.

6 Discussion

6.1 Computer-vision findings and generalization

The parsing results show why model selection should follow the intended generalization axis rather than a single in-domain metric. DeepLabV3+ is strongest on the QC-clean standard split, SegFormer-B2 has the best character-disjoint direction overlap and the smallest decline between protocols, and U-Net remains strongest for strict endpoint localization. Dense directional regions and sparse endpoint targets therefore favour different architectures. More broadly, the gap between standard and character-disjoint performance demonstrates that limited-class dense segmentation can appear nearly saturated while transfer to unseen structural compositions remains materially weaker. Reporting only an identity-overlapping split would conceal this distinction.

The overlapping output contract addresses a limitation of ordinary mutually-exclusive segmentation rather than proposing another character classifier. At adhered and crossing regions, independent sigmoid channels preserve multiple valid directional memberships. This representation is useful wherever layered document primitives share foreground support and the downstream task requires semantic evidence rather than a foreground-only partition. The qualitative cases and channel-level metrics are consequently part of the scientific result: a union mask alone cannot reveal a direction-label swap or an endpoint failure.

Registration is likewise part of the measurement design rather than a neutral preprocessing step. Without alignment, the tested global geometric changes dominate the score. Relative to no alignment, the production range substantially reduces nuisance sensitivity while retaining penalties for structural intervention. The wider search does not improve the trade-off, showing that additional transform capacity is not automatically beneficial when discrepancies are diagnostically meaningful.

These results support a deliberately limited claim. Translation is recovered well, but rotation and especially scale retain non-negligible penalties. Scale-up and compound summaries are conditional on clipping-safe references, with invalid fractions of 49.0% and 54.0%. Because clipping safety depends on glyph extent, the reported means are conditional estimates and do not establish invariance across the full library or even across every transform in the nominal search range. The appropriate conclusion is that constrained registration reduces the measured nuisance sensitivity relative to no alignment while preserving more structural discrepancy than a freely deformable comparison would be expected to preserve.

The inactive-channel audit exposes a second general measurement issue. Both-empty semantic channels can receive perfect overlap credit despite carrying no observed structure. The mean effect is small, but individual corrections are large and the coverage-aware audit has stronger association with human ratings. Multi-label document systems should therefore report label coverage and audit empty-channel semantics rather than relying on a macro average whose meaning changes with the active label set.

6.2 Assessment validity and diagnostic implications

The human study separates three levels of evidence. The production score was frozen before the rating analysis and has a modest positive association with mean structural ratings. The coverage-aware audit was also defined from an explicit semantic concern and has a stronger association. ASDS gives the largest observed association, but its feature families and weights were finalized after the original ratings were available. It would therefore be incorrect to present $\rho = 0.556$ as an external or prospectively confirmed result.

We instead retain the original score results, disclose the complete low-capacity candidate search, report component ablations, and distinguish the full-sample development estimate from character-grouped out-of-fold internal validation. The latter reduces weight-selection optimism but cannot remove the fact that the representation itself was proposed after inspecting the same study. ASDS should remain frozen in any subsequent collection, where its performance can be tested on independently recruited writers, characters, and raters without another feature search.

ASDS also clarifies, rather than replaces, the role of the six-channel parser. Its polar occupancy, 3×3 occupancy, and projection profiles use only the union silhouette. On clean direct-digital input, a parser is not mathematically necessary merely to obtain that silhouette. The paired direct-ink ablation confirms the point empirically: parsed-union and direct-ink ASDS differ by only 0.0004 in Spearman correlation, and the paired interval includes small advantages in either direction. We find no evidence that parsing improves the scalar ASDS association through denoising or normalization in this modality.

This negative result prevents circular justification. The parser remains necessary for the separate claims that the system parses overlapping direction families, measures channel-wise agreement, detects missing or extra directional support, and represents endpoint relations. A direction-label swap can preserve the same union silhouette and hence the same ASDS while changing all of those outputs. Applications that request only a scalar silhouette score from an already clean raster may compute direct-ink ASDS; applications that require localized structural explanation still require the parser and its independently evaluated output contract.

Human agreement further bounds the interpretation. Average-rating ICC is higher than single-rating ICC, supporting aggregation of multiple judgments, but the three-rater pool is small and heterogeneous. One rater reported limited calligraphy learning/teaching experience, and the group is not a panel of senior calligraphy experts. The evidence supports the terms "human-audited structural similarity" and "reference-conditioned agreement", not expert aesthetic grading. A follow-up study should recruit experienced calligraphy educators under a fixed background questionnaire and test the frozen score without replacing or relabelling the present cohort.

The diagnostic analysis also argues for separating evidence from language. Low exact-region accuracy is driven mainly by a single-cell metric applied to multi-cell perturbations, followed by genuine direction-attribution failures. A region-set or pixel-overlap target is therefore a more appropriate next evaluation than tuning prose rules on the same cases. Optional language generation should remain downstream of the structured evidence contract: fluent wording cannot correct an erroneous direction, endpoint relation, or spatial localization.

6.3 Limitations and threats to validity

The segmentation corpus contains only 40 fully labelled character identities. It was collected through a commissioned, calligraphy-trained writing procedure, but the recovered records do not preserve reliable writer identity or standardized contributor credentials. Character-disjoint evaluation is therefore not writer-disjoint evaluation, and both parsing protocols originate from one recovered source. Semantic QC removes known mismatches and exact duplicate leakage, but unobserved collection-specific regularities may remain.

Controlled perturbations operate on cached reference masks and intentionally isolate registration and score behaviour from parser error. An end-to-end failure can combine segmentation, registration, and comparison errors. Furthermore, the scale-up and compound analyses condition on clipping-safe references. They should not be generalized to a full-library transform invariance claim, and further tuning on the same perturbation suite would risk benchmark-specific overfitting.

The reference library includes two style categories, only seven same-character cross-style pairs, and no same-style different-instance reference pair. The human development cohort contains 150 pairs and three raters drawn from the recovered project corpus. Character-grouped resampling limits dependence across character identities but does not establish generalization to new writers, collections, or evaluator populations. ASDS can also ignore a direction-label swap that preserves union ink, so it must remain complementary to channel-level evidence.

Finally, the target input is clean direct digital writing. The study does not test paper texture, illumination, perspective, or phone photography. SegFormer-B2 benefits from a GPU for interactive latency, while cached references, registration, ASDS, deterministic evidence, and language-API realization can be served without a training-class accelerator. Deployment cost, distillation, and quantized edge inference are engineering questions outside the present scientific claims.

7 Conclusion

We formulated low-resource Chinese calligraphy assessment as a computer-vision problem in which overlapping document structures must be parsed, globally aligned without erasing diagnostically meaningful differences, and compared under an explicit human-validity claim. The six-channel representation preserves independent directional memberships at crossings and endpoint evidence that a foreground-only partition cannot provide. Across three fixed seeds, DeepLabV3+ is strongest on the QC-clean standard split, SegFormer-B2 gives the best character-disjoint direction result, and U-Net gives the best strict endpoint result. The resulting architecture-by-target interaction and the standard-to-character-disjoint gap show why low-resource dense parsing should be evaluated beyond an identity-overlapping split.

Controlled perturbations and paired ablations show that constrained registration substantially reduces nuisance sensitivity relative to no alignment while retaining penalties for local structural changes. This is a relative robustness result, not transform invariance: rotation and scale remain imperfectly compensated, and scale-related summaries are conditional on clipping-safe references. The inactive-channel audit further demonstrates that the semantics of a macro overlap score require inspection when the active label set varies by sample.

The human-rating analysis provides a similarly bounded conclusion. The deployed formula has a modest positive association with blinded structural ratings, and the coverage-aware audit is stronger. ASDS increases development-stage Spearman correlation from 0.297 to 0.556 and retains 0.540 under character-grouped out-of-fold prediction, but these are retrospective development and internal-validation results rather than external confirmation. Parsed-union and direct-ink ASDS are nearly equivalent on clean direct-digital inputs ($\Delta\rho = 0.0004$, 95% CI -0.0101–0.0103); the scalar silhouette descriptor therefore cannot be used as the justification for the parser. The parser's independent value lies in overlapping direction segmentation, channel-aware comparison, endpoint relations, and localized missing/extra evidence.

The principal contribution to document analysis is thus an auditable evaluation framework for limited-label visual assessment: semantic data QC, standard and unseen-class parsing protocols, discrepancy-preserving registration tests, score-semantics audits, blinded human association, and explicit separation of retrospective development from confirmation. Within its direct-digital scope, the system supports reference-conditioned structural analysis rather than universal aesthetic grading. The same design principles are applicable to other low-resource documents and line-art domains in which intersecting components, nuisance geometry, and localized explanation must be handled jointly.

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Declarations

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Conflict of interest

The authors declare no competing interests.

Ethics approval and consent to participate

The structural-similarity study involved adult volunteers, stored ratings under evaluator codes, and did not collect sensitive personal information. Before submission, the authors will insert the institution-specific ethics approval, exemption, or review statement and the applicable consent wording. No approval or exemption is claimed in this draft.

Consent for publication

Not applicable, subject to confirmation by the authors before submission.

Data availability

Code and data are maintained as separate releases. A versioned data archive is being prepared for the project-owned QC-clean manifests and split contracts, the blinded 150-pair rating table, perturbation specifications, derived statistics, and integrity hashes. Its permanent repository identifier will be inserted before submission. The archive will not redistribute Calli-Tongji source images, which remain subject to the original CC-BY-NC-4.0 licence, and project-collected RGB/stroke-mask assets will be released only to the extent permitted by participant consent and institutional data-governance review. Where an image cannot be redistributed, the archive will provide stable identifiers, hashes, provenance, and regeneration or access instructions.

Code availability

Source code, frozen experiment contracts, analysis scripts, and non-restricted reproducibility artifacts are maintained independently from the data archive at https://github.com/LiuXiaofan-0321/OneStroke2026. If anonymous review is required, an identity-free archival copy will be supplied through the submission system.

Author contributions

X.L.: Conceptualization, methodology, software, formal analysis, visualization, writing—original draft, project administration. R.Z.: Data curation, methodology, software, validation, writing—review and editing. Y.F.: Software, system integration, validation, writing—review and editing. All authors contributed to the investigation, approved the submitted manuscript, and accept responsibility for the work.

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